# CS1502
# Formal Methods in Computer Science
# Syllabus

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Please see updated information on the Canvas

# Instructor

- **Name**: Thumrongsak Kosiyatrakul
  - Feel free to call me Tan, Dr. Tan, or Professor Tan



- **Email**: `tkosiyat@pitt.edu`
  - Please start the subject line with `[CS1502]` ...
- **Office Location**: 6215 SENSQ
- **Office Hours**: (TuTh) 9:30 AM to 11:30 AM (EST)
- **Office Hours URL**: Check Canvas for the updated URL

# Teaching Assistant

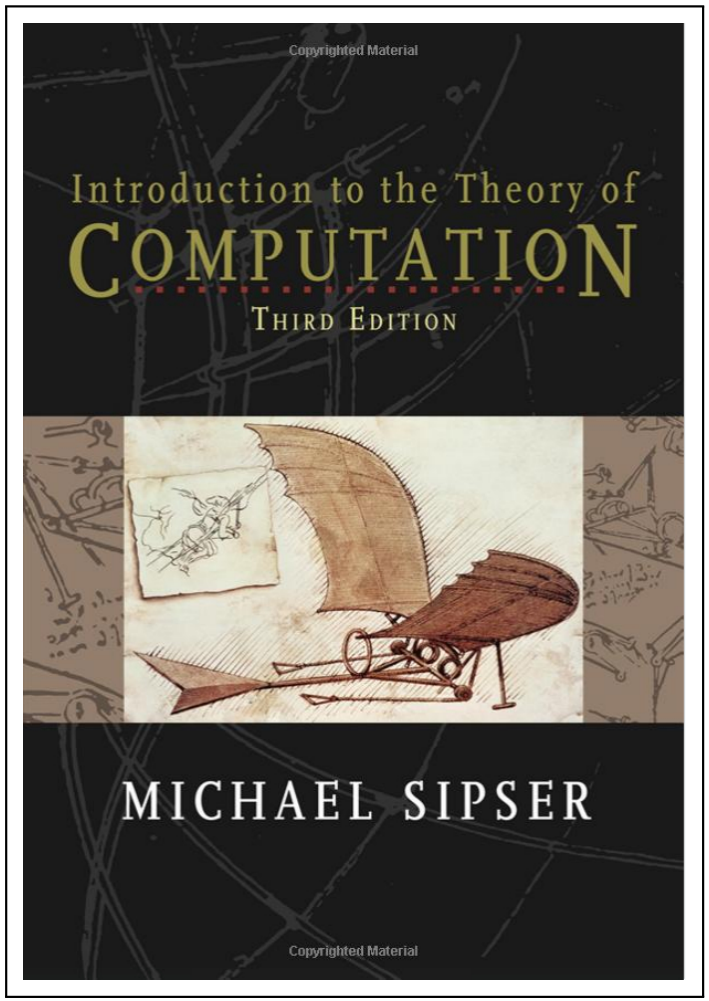
- **Name**: TBA
- **Email**: TBA
- **Office Location**: TBA
- **Office Hours**: TBA
- **Office Hours URL**: TBA

# Meeting Information

- **Lecture**:
  - **Date/Time**: According to your section
  - **Location (URL)**: Check Canvas for the updated URL
- For Spring 2021:
  - Pre-recorded lectures will be provided
    - Almost like the flip classroom style
    - Watch them in advance is recommended
    - All slides are provided with the lecture
    - Available under **Modules**
  - Lecture sessions will be used for summaries, quick overview, practice questions, and Q&A
    - Available under **Panopto**

# Textbook

- **Authors**: Michael Sipser
- **Name**: Introduction to the Theory of Computation
- **Edition**: $2^{\text{nd}}$ or $3^{\text{rd}}$ edition



# Outcome Measurement

- **Exams**: 62% (2 exams 31% each)
- **Assignments**: 31% ($\approx 7 - 8$ written assignments)
  - No late submissions will be accepted
  - Upload your assignment to the Canvas
    - `.pdf` format is preferred
    - Handwritten, take pictures, and convert them into a single `pdf` file is okay
- **Take-Home Quizzes**: 5% ($\approx 10 - 12$ quizzes)
- **Assessment Tests**: 2% (2 tests)

| $\geq 95\%$ | $\geq 90\%$ | $\geq 89\%$ | $\geq 88\%$ | $\geq 80\%$ | $\geq 79\%$ | $\geq 78\%$ | $\geq 70\%$ |
|---|---|---|---|---|---|---|---|
| A+ | A | A- | B+ | B | B- | C+ | C |

# Course Web Site

- **Canvas** will be our LMS
  - Slides
  - Recorded lectures
  - Exams
  - Assignments
  - Take-home quizzes
  - Assessment Tests
- Always start from the Canvas front page for updated URLs

# Outline Schedule

- **Chapter 1**: Finite Automata
- **Chapter 3**: Turing Machines
- Midterm Exam
- **Chapter 4**: Decidability
- **Chapter 5**: Reducibility
- **Chapter 7**: Time Complexity
- Final Exam (last day of lecture)

# Other Information

- Academic Integrity
- Disability Services
- Statement on Classroom Recording

- There are some problems that cannot be solved
  - Given a polynomial with multiple variables:

    $$6x^3yz^2 + 3xy^2 - x^3 - 10$$

    Does it has an integral root?
  - Given a set of dominoes:

    $$\left\{ \left[\frac{\texttt{b}}{\texttt{ca}}\right], \left[\frac{\texttt{a}}{\texttt{ab}}\right], \left[\frac{\texttt{ca}}{\texttt{a}}\right], \left[\frac{\texttt{abc}}{\texttt{c}}\right] \right\}$$

    Is it possible to make a horizontal line of one or more dominoes, with duplicates allowed, so that the string obtained by reading across the top halves **matches** the one obtained by reading across the bottom?
    - Post Correspondence Problem

- There are infinite number of algorithms
  - How to **prove** that none of them can be used to solve a problem?
- Problem
  - No mathematical representation that can be used to express all algorithms
- Computational Models:
  - Finite Automata:
    - Capture a subset of algorithms
  - Turing Machine:
    - Capture all algorithms
- Learn about limitations of computational models (limitations of algorithms) and how to formally verify their limitations

- There are some problem that take too long to solve
  - Unrealistically solvable
- Modified Travel Salesman Problem
  - Shortest path to visit 1000 cities?
    - There are 1000! possible path
    - $1000! \approx 4 \times 10^{2567}$
    - Suppose you can check the distance of each path in 1 ns, it will take $\approx 4 \times 10^{2558}$ seconds
    - $\approx 4 \times 10^{2556}$ minutes
    - $\approx 4 \times 10^{2554}$ hours
    - $\approx 4 \times 10^{2552}$ days
    - $\approx 4 \times 10^{2549}$ years
- Do we have a faster algorithm?
  - Maybe not
- Need to learn how to spot these kind of problems in advance

# Proofs

- To **prove** that a given statement is true is to demonstrate the truth of the given statement
- A typical step in a **Proof** is to derive a true statement using principles of logical reasoning from one of the following:
  - Assumptions
  - Hyptheses
  - Statements that have been derived previously
  - General accepted facts
  - Definitions

# Example 1

- Let $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$
- **Definition** (simple one): An **integer** is a number in $\mathbb{Z}$
- **Formally**: $n$ is an integer if an only if $n \in \mathbb{Z}$
  - Let $P(n)$ be "$n$ is an integer"
  - Let $Q(n)$ be "$n \in \mathbb{Z}$"
  - The above statement is $P(n) \leftrightarrow Q(n)$
- **Prove** that "$5$ is an integer"
  - The statement "$5$ is an integer" is $P(5)$
  - We need to show that $P(5)$ is true
  - Since $P(5) \leftrightarrow Q(5)$, if $Q(5)$ is true, $P(5)$ is true
  - So, we need to show that $Q(5)$ is true
  - $Q(5)$ is "$5 \in \mathbb{Z}$"
  - Since $5 \in \mathbb{Z}$, $Q(5)$ is true
  - Therefore, the statement "$5$ is an integer" is true

# Example 2

- **Definition**: An even number $n$ is an integer of the form $n = 2k$ where $k$ is an integer
- **Formally**: $n$ is even if an only if $n$ is an integer and $n = 2k$ and $k$ is an integer
- **Prove** that "$12$ is even"
  - We need to show the following:
    - $12$ is an integer (done)
    - $12 = 2k \rightsquigarrow k = 6$
    - $6$ is an integer (done)
  - Therefore, "$12$ is even is true"
- **Definition**: An odd number $n$ is an integer of the form $n = 2k + 1$ where $k$ is an integer

# Example 3

**Prove**

For every two integers $a$ and $b$, if $a$ and $b$ are odd, then $a \times b$ is odd.

- The above statement is in the form $P \rightarrow Q$ where
  - $P$ is "$a$ and $b$ are odd"
  - $Q$ is "$a \times b$ is odd"
- Recall the truth table of $P \rightarrow Q$

| $P$ | $Q$ | $P \rightarrow Q$ |
|---|---|---|
| T | T | T |
| T | F | F |
| F | T | T |
| F | F | T |

- So, to show that $P \rightarrow Q$ is true, we need to show that if $P$ is true, $Q$ must be true
- Need to assume $P$ to be true and show that $Q$ is also true

# Example 3

**Prove**

For every two integers $a$ and $b$, if $a$ and $b$ are odd, then $a \times b$ is odd.

- Initial Assumptions:
  - $a$ and $b$ are integers.
  - $a$ and $b$ are odd.
  - **Note**: $a$ and $b$ can be **any** integers that are odd.
- Statement to Derive:
  - $a \times b$ is odd.
- Fact:
  - If $x$ is an integer that is odd, there exists an integer $k$ such that $x = 2k + 1$.
- From the initial assumptions that $a$ and $b$ are odd integers, we can derive:
  - There exists two integers $i$ and $j$ such that $a = 2i + 1$ and $b = 2j + 1$.

# Example 3

**Prove**

For every two integers $a$ and $b$, if $a$ and $b$ are odd, then $a \times b$ is odd.

- Let $a = 2i + 1$ and $b = 2j + 1$:

$$\begin{aligned} a \times b &= (2i + 1) \times (2j + 1) \\ &= 4ij + 2i + 2j + 1 \\ &= 2(2ij + i + j) + 1 \end{aligned}$$

- Thus, $a \times b$ is an odd integer since there exists an integer $k$ $(2ij + i + j)$ such that $a \times b = 2k + 1$.
- **Note**: By showing a couple of test cases is not a proof. In the above proof, we assume that $a$ and $b$ can be any odd integers.

# Proof by Contradiction

- A statement $P$ is logically equivalent to $\neg P \rightarrow$ False

| $P$ | $\neg P$ | $\neg P \rightarrow$ False |
|---|---|---|
| False | True | False |
| True | False | True |

- Thus, to show that the statement $P$ is True, we can show that the statement $\neg P \rightarrow$ False is true.
  - In other words, assume that $P$ is False and derive a statement that **contradict** with an initial assumption, an hypotheses, a statement that have been derived previously, or another generally accepted fact.

# Example 4

**Prove**

For every integer $p$, if $p^2$ is even, $p$ is even.

- **Formally**: "$p^2$ is even" $\rightarrow$ "$p$ is even"
- Assume that the above statement is false
- Recall that

$$\begin{aligned}\neg(p^2 \text{ is even} \rightarrow p \text{ is even}) &\equiv \neg(\neg(p^2 \text{ is even}) \vee p \text{ is even}) \\ &\equiv p^2 \text{ is even} \wedge \neg(p \text{ is even}) \\ &\equiv p^2 \text{ is even} \wedge p \text{ is odd}\end{aligned}$$

- Note that
  - The statement "$p$ is odd" is either true or false
  - If "$p$ is odd" is false, the above statement is false
  - If "$p$ is odd" is true, "$p^2$ is even" is false
  - In other word, the above statement implies false
- Therefore, "if $p^2$ is even, $p$ is even" is true

# Example 5

- **Definition**: A **rational** number is a number such as $\frac{-3}{7}$ that can be expressed as the quotient or fraction $\frac{p}{q}$ of two integers, a numerator $p$ and a non-zero denominator $q$
- Every rational number may be expressed in a unique way as an **irreducible fraction** $\frac{a}{b}$, where $a$ and $b$ are coprime integers and $b > 0$
  - No common integer factor greater than $1$ between $a$ and $b$
  - Canonical form

# Proof by Contradiction

**Prove**

$\sqrt{2}$ is an irrational number.

- Fact: An irrational number is any real number that cannot be expressed as a ratio of integers.
  - In other words, there is not exists two integers $m$ and $n$ such that $\frac{m}{n} = \sqrt{2}$.
- Let $P$ be the statement "$\sqrt{2}$ is an irrational number".
- Thus $\neg P$ is "$\sqrt{2}$ is a rational number".
  - In other words, there exists two integers $m$ and $n$ such that $\frac{m}{n} = \sqrt{2}$.
- Fact: For any ration $\frac{m}{n}$ we can obtain $\frac{m}{n} = \frac{p}{q}$ for some positive integers $p$ and $q$ with no common factors.

# Proof by Contradiction

**Prove**

$\sqrt{2}$ is an irrational number.

- Let $\frac{p}{q} = \sqrt{2}$ for some positive integers $p$ and $q$ with no common factor.
  - **Note** that the above statement says that $\sqrt{2}$ is a rational number
- Let's try to derive some statements

$$\frac{p}{q} = \sqrt{2}$$

$$\left(\frac{p}{q}\right)^2 = (\sqrt{2})^2$$

$$\frac{p^2}{q^2} = 2$$

$$p^2 = 2q^2$$

# Proof by Contradiction

**Prove**

$\sqrt{2}$ is an irrational number.

- Since $p^2 = 2q^2$, $p^2$ is an even number.
  - In other words, $p$ is an even number or there exists an integer $r$ such that $p = 2r$.
- Keep deriving:

$$\begin{aligned} p^2 &= 2q^2 \\ (2r)^2 &= 2q^2 \\ 4r^2 &= 2q^2 \\ 2r^2 &= q^2 \end{aligned}$$

- Since $q^2 = 2r^2$, $q^2$ is an even number ($q$ is an even number)

# Proof by Contradiction

**Prove**

$\sqrt{2}$ is an irrational number.

- Earlier we stated that $\frac{p}{q} = \sqrt{2}$ for some positive integers $p$ and $q$ with no common factor.
- But we show that if $\sqrt{2}$ is a rational number both $p$ and $q$ are even.
- Therefore $p$ and $q$ have a common factor which is **False** according to our assumption.
- So, we get $\neg P \rightarrow$ False which is equivalent to $P$.
- Thus, $\sqrt{2}$ is an irrational number.